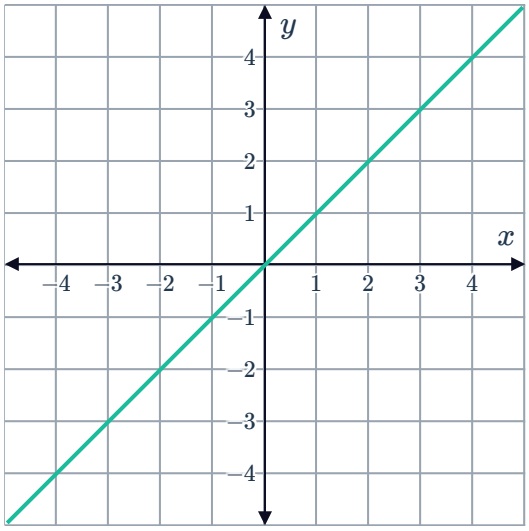
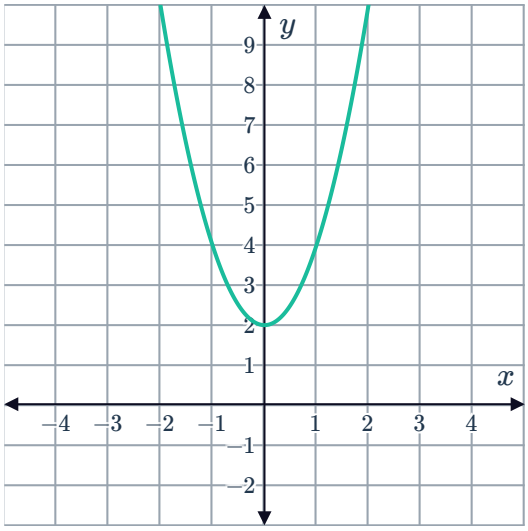
Equations of lines

1 Determine whether the following graphs are linear:

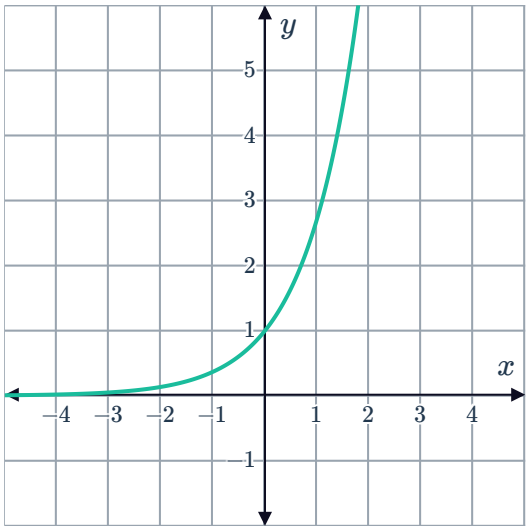
a



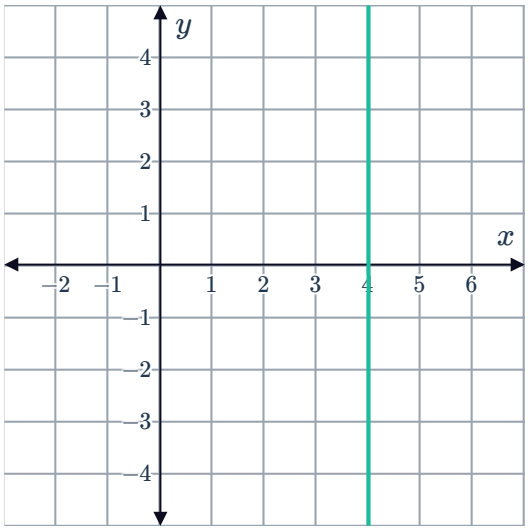
b



c

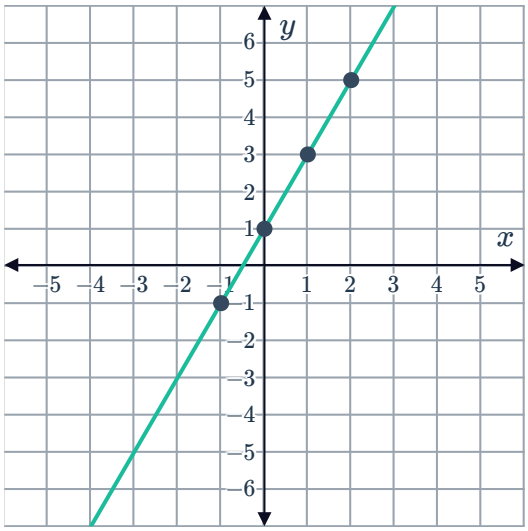


d



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2 Consider the following graph:



a Complete the table of values for the points shown on the line.

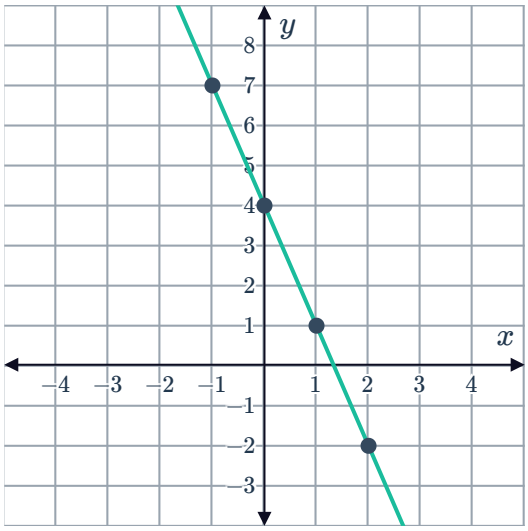
x	-1	0	1	2
y				

b State the values of the gradient m, and the y-intercept, c.

c Given that the linear relationship is in the form $y = mx + c$, write the equation of the graph.

d Hence find the coordinates for the point on the line when $x = 21$.

3 Consider the following graph:



a Complete the table of values for the points shown on the line.

x	-1	0	1	2
y				

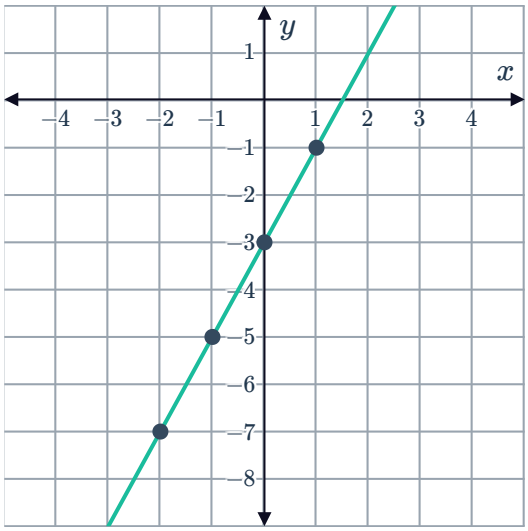
b State the values of the gradient m, and the y-intercept, c.

c Hence write the linear equation for the graph.

d Find the coordinates for the point on the line where $x = 27$.

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4 Consider the following graph:



a Complete the table of values for the points shown on the line:

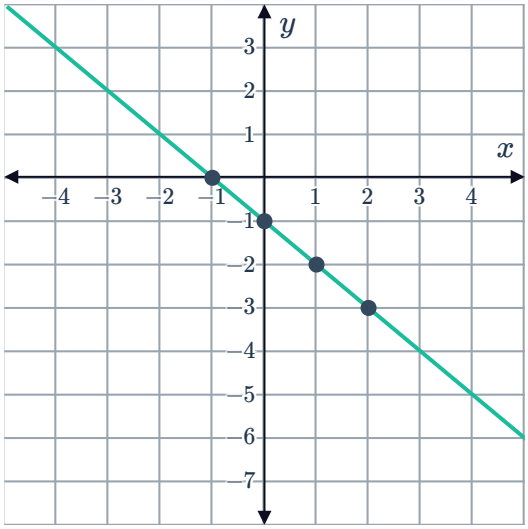
x	-2	-1	0	1
y				

b State the values of the gradient m, and the y-intercept, c.

c Hence write the equation for the graph.

d Find the value of y when $x = 40$.

5 Consider the following graph:



a Complete the table of values for the points shown on the line:

x	-1	0	1	2
y				

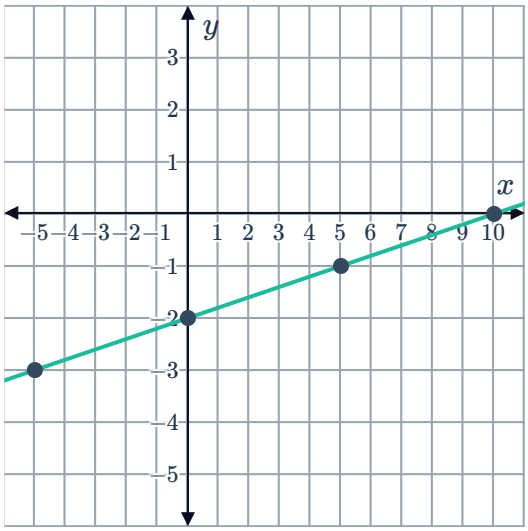
b State the values of the gradient m, and the y-intercept, c.

c Hence write the equation for the graph.

d Find the value of y when $x = 19$.

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6 Consider the following graph:



a Complete the table of values for the points shown on the line:

x	-5	0	5	10
y				

b When x increases by 5, by how much does y increase?

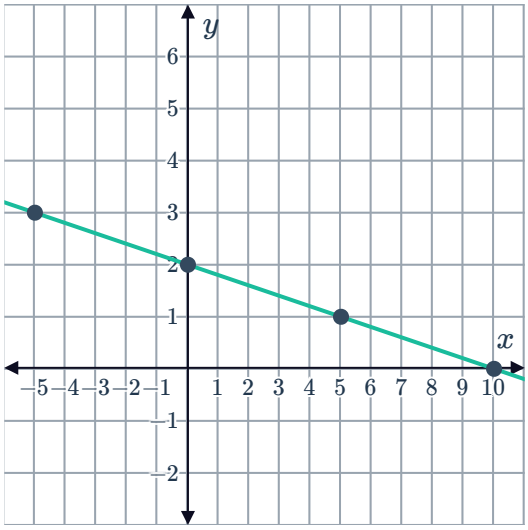
c Hence when x increases by 1, by how much does y increase?

d State the values of the gradient m, and the y-intercept, c.

e Hence write the equation for the graph.

f Find the coordinates for the point on the line where $x = 35$.

7 Consider the following graph:



a Complete the table of values for the points shown on the line:

x	-5	0	5	10
y				

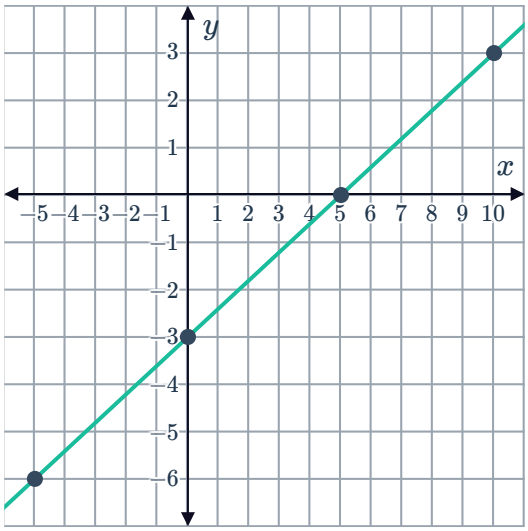
b State the values of the gradient m, and the y-intercept, c.

c Hence write the equation for the graph.

d Find the coordinates for the point on the line where $x = -45$.

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8 Consider the following graph:



a Complete the table of values for the points shown on the line:

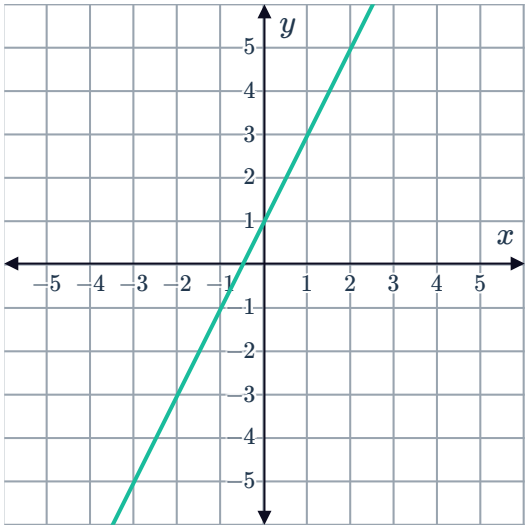
x	-5	0	5	10
y				

b State the values of the gradient m, and the y-intercept, c.

c Hence write the equation for the graph.

d Find the coordinates for the point on the line where $x = 45$.

9 Consider the following graph:



a Complete the table of values for the line shown in the graph:

x	0	1
y		

b As x increases by 1, by how much does y increase?

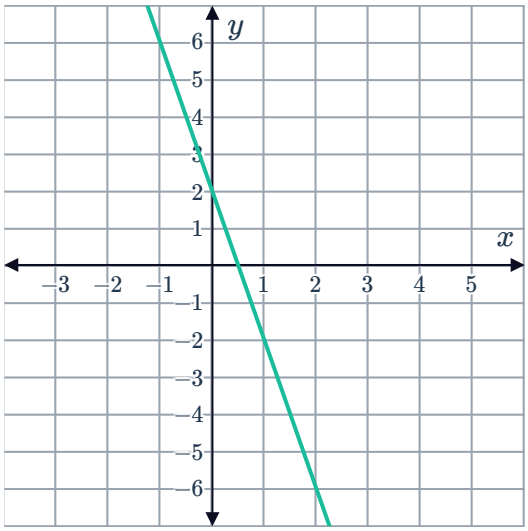
c State the values of the gradient m, and the y-intercept, c.

d Hence write the equation of the line.

e Find the y-value of the point on the line where $x = 22$.

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10 Consider the following graph:



a Complete the table of values for the line shown in the graph:

x	0	1
y		

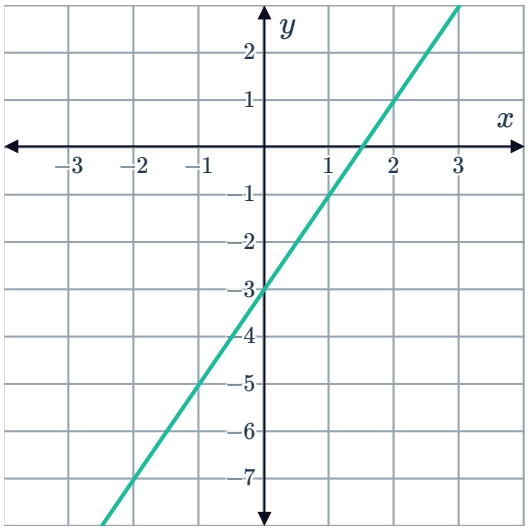
b As x increases by 1, by how much does y decrease?

c State the values of the gradient m, and the y-intercept, c.

d Hence write the equation of the line.

e Find the y-value of the point on the line where $x = 27$.

11 Consider the following graph:



a As x increases by 1, by how much does y increase?

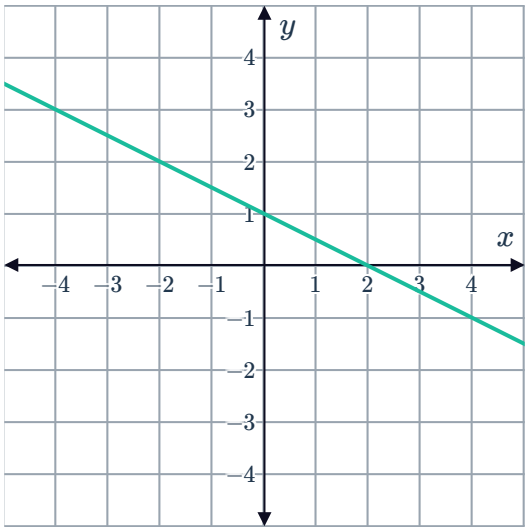
b State the values of the gradient m, and the y-intercept, c.

c Hence write the equation of the line.

d Find the coordinates for the point on the line where $x = 50$.

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12 Consider the following graph:



a As x increases by 2, by how much does y decrease?

b Hence as x increases by 1, by how much does y decrease?

c State the values of the gradient m, and the y-intercept, c.

d Hence write the equation of the line.

e Find the coordinates for the point on the line where $x = -14$.

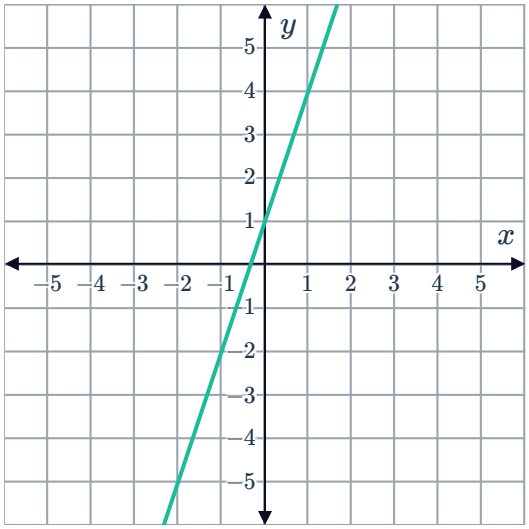
13 For each of the following graphs:

i State the coordinates of the y-intercept.

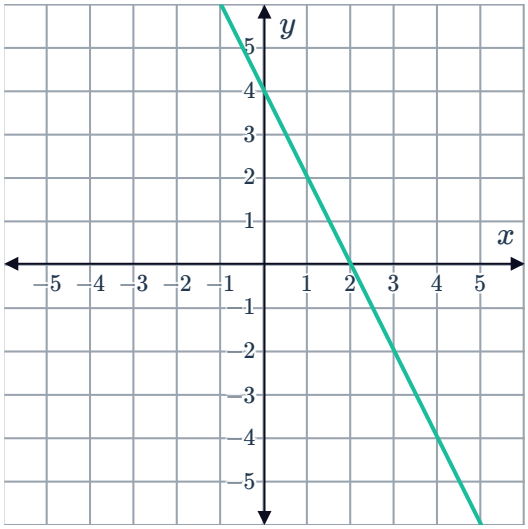
ii Find the gradient of the line.

iii Write the linear equation expressing the relationship between x and y.

a

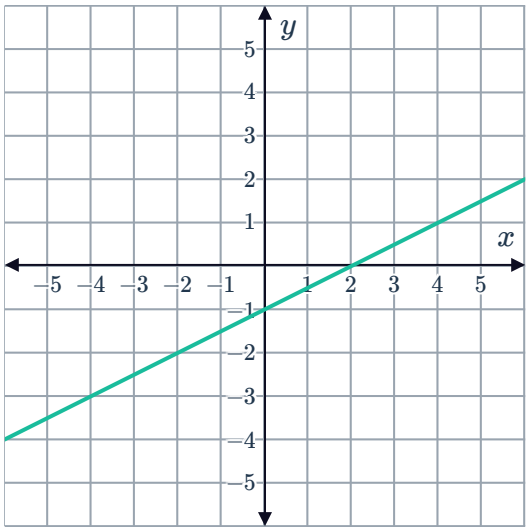


b

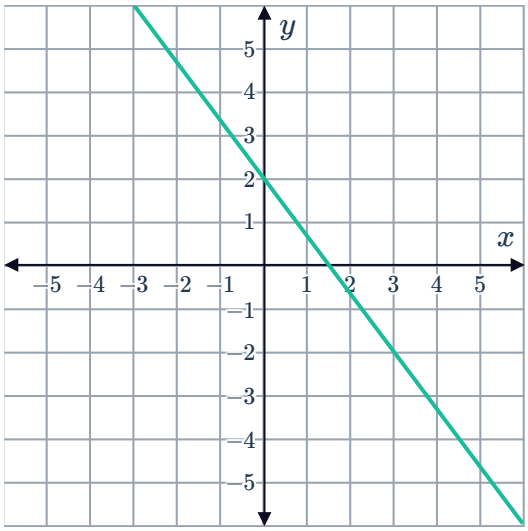


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c

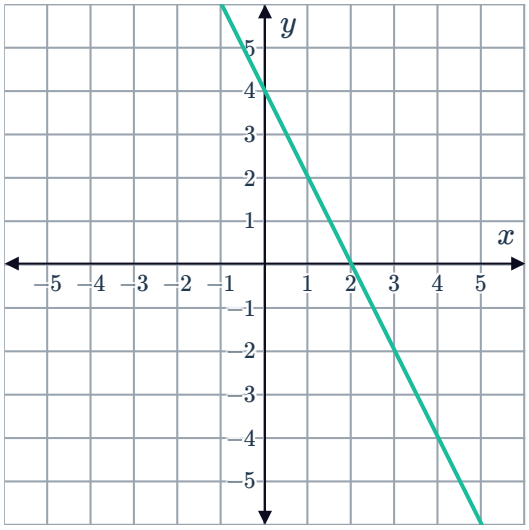


d

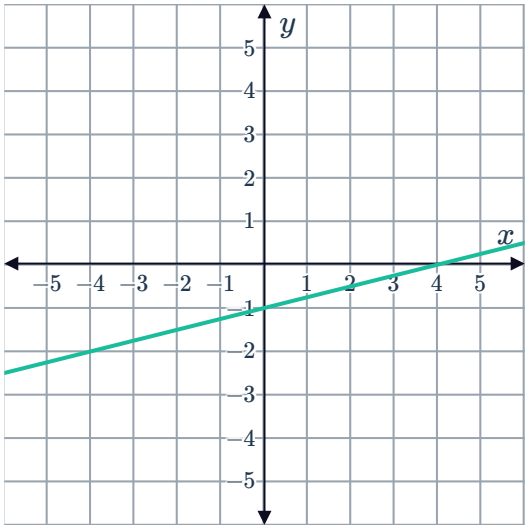


14 Find the equation expressing the relationship between x and y, for each of the following graphs:

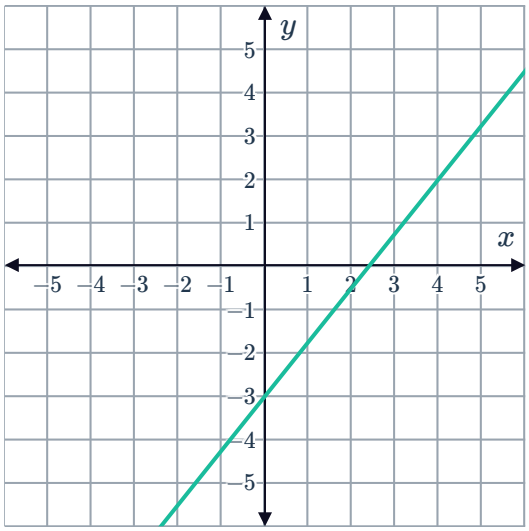
a



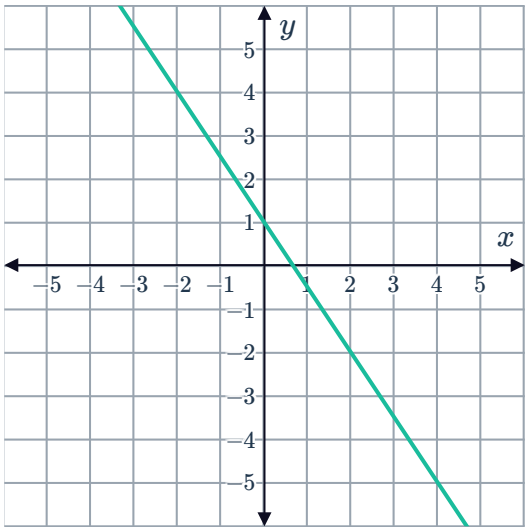
b



c

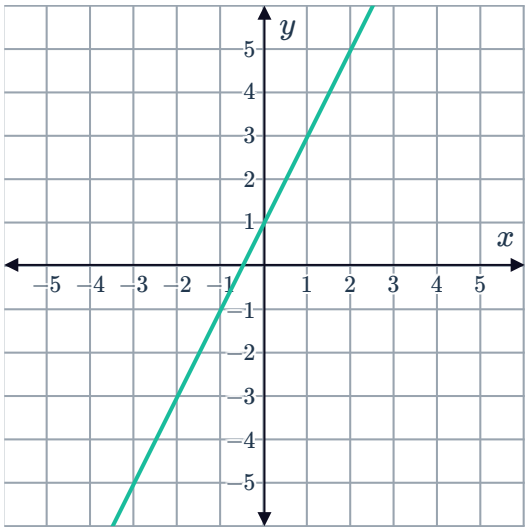


d



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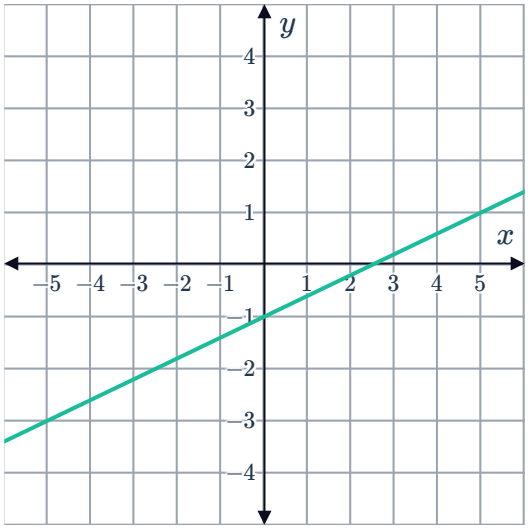
15 Consider the following graph:



a Find the linear equation expressing the relationship between x and y.

b Hence find the coordinates for the point on the line where $x = 22$.

16 Consider the following graph:



a Find the linear equation expressing the relationship between x and y.

b Find the coordinates for the point on the line where $x = 25$.

17 Find the equation of the line, given the following information:

a The line has y-intercept of 4, and gradient of -3.

b The line has y-intercept of -2, and gradient of 9.

c The line contains the point $(0, 7)$ and has gradient of -1.

d The line contains the origin and has gradient of $\frac{3}{4}$.

e The line contains the origin and the point $(1, 3)$.

f The line contains the origin and the point $(1, -5)$.

g The line contains the points $(0, -5)$ and $(1, 1)$.

h The line contains the points $(0, 2)$ and $(2, 6)$.

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